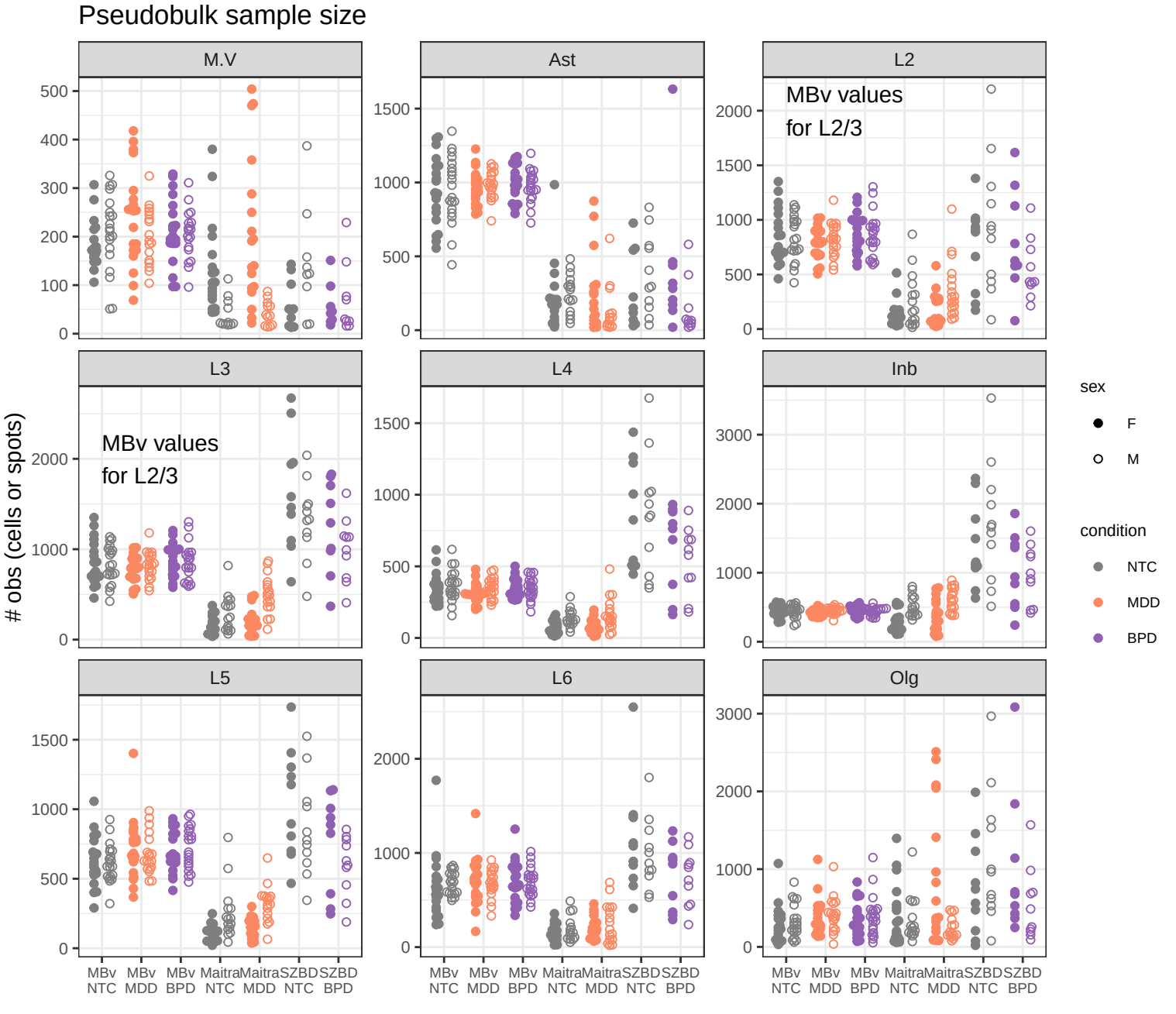
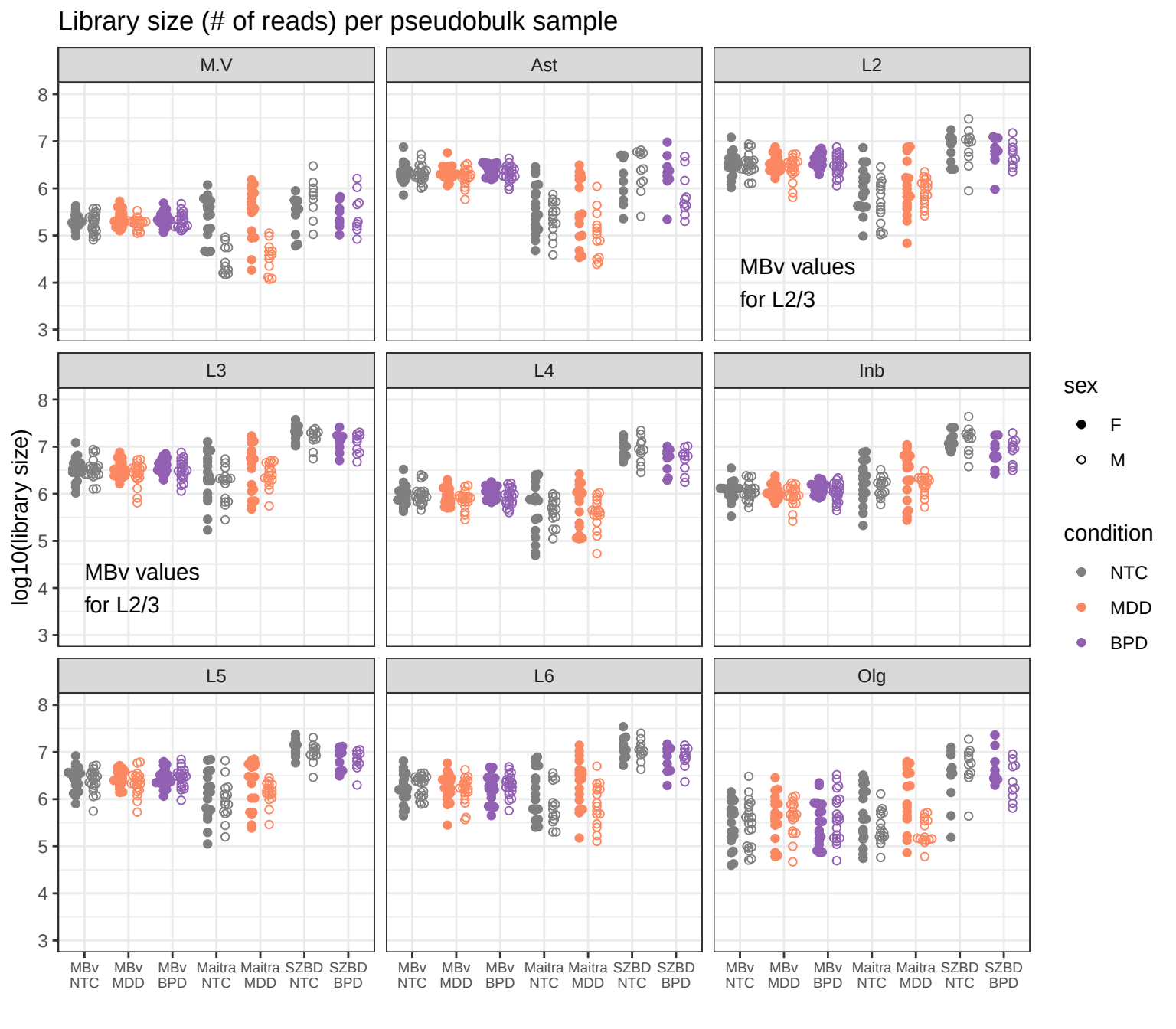
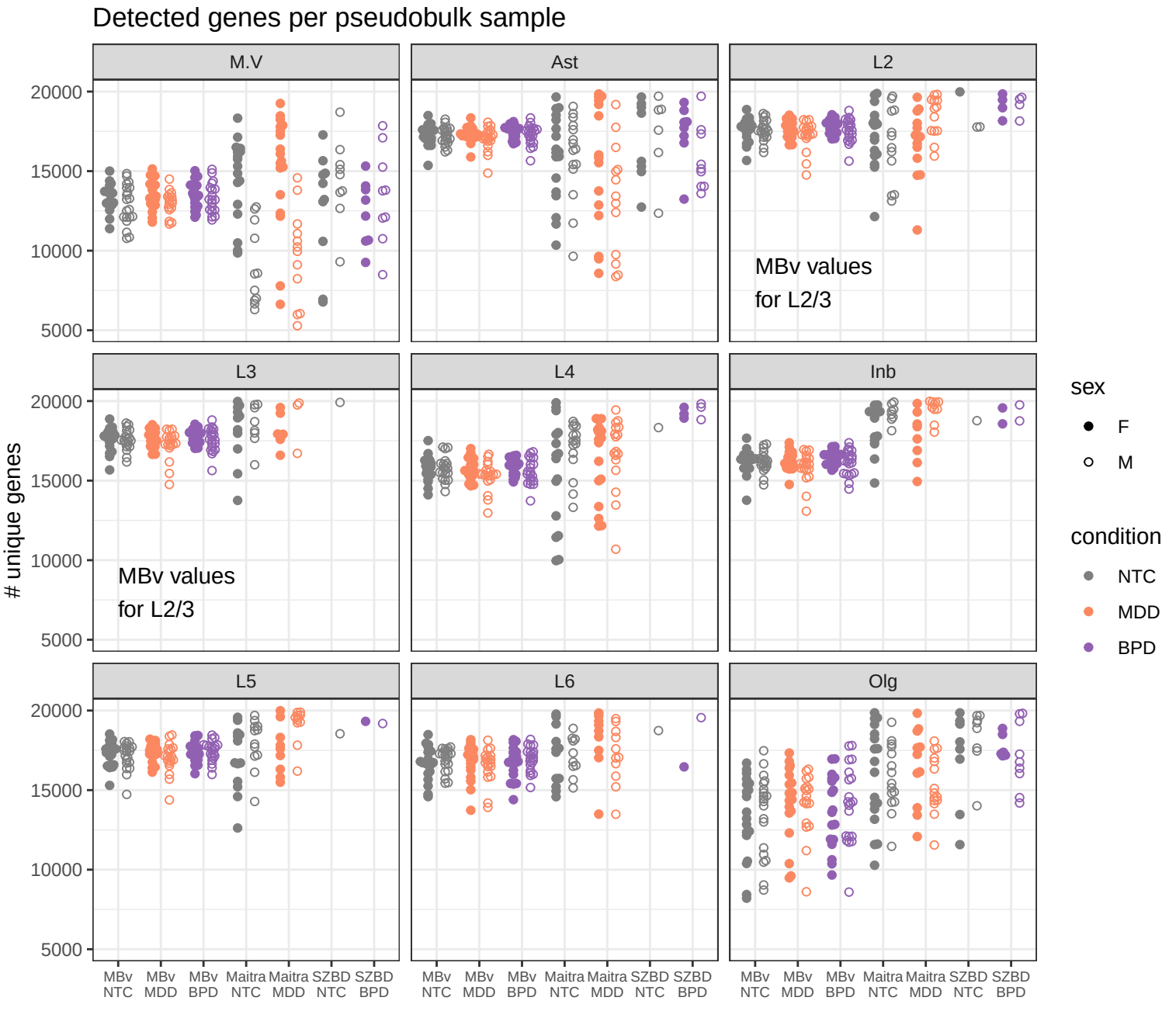
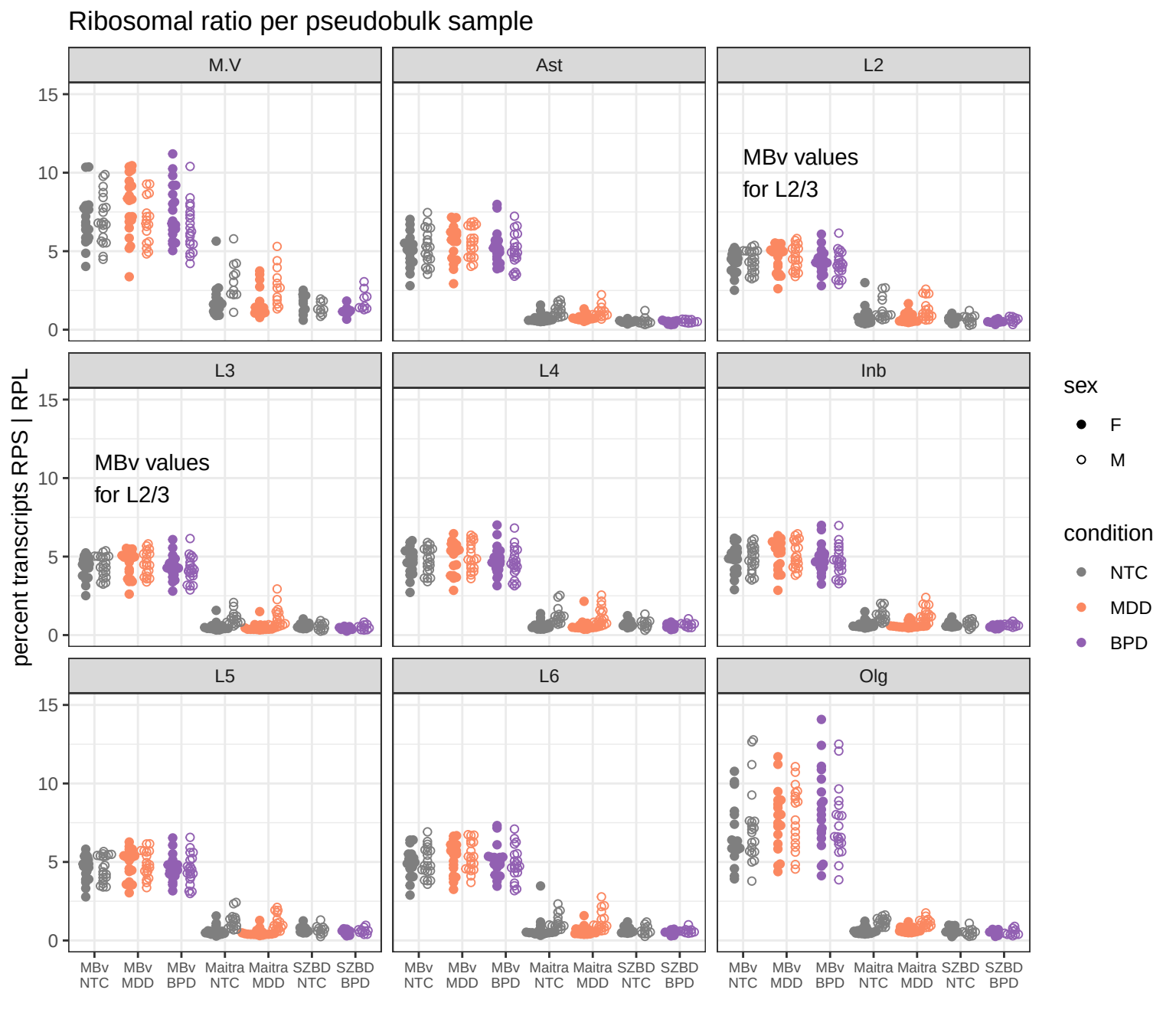
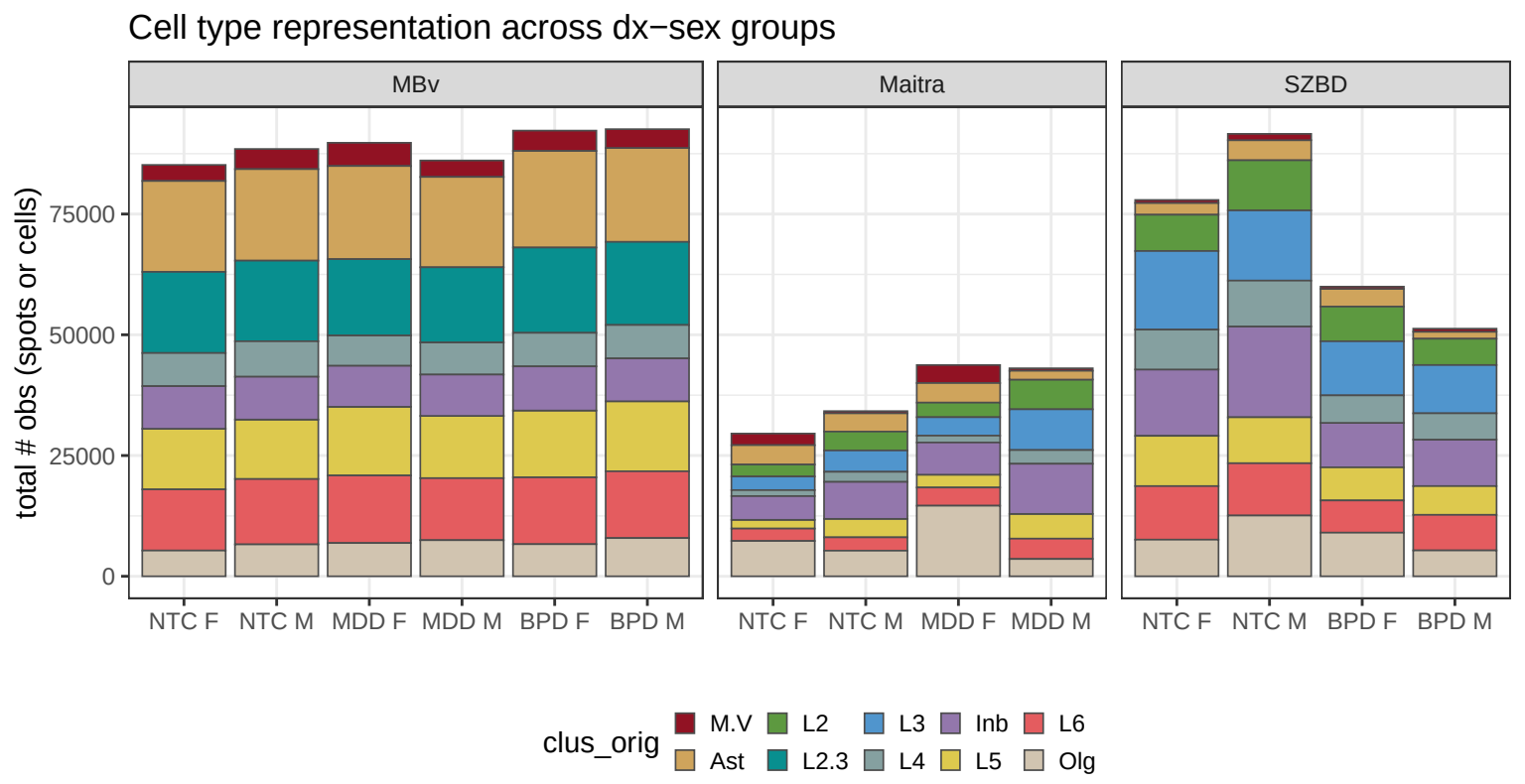
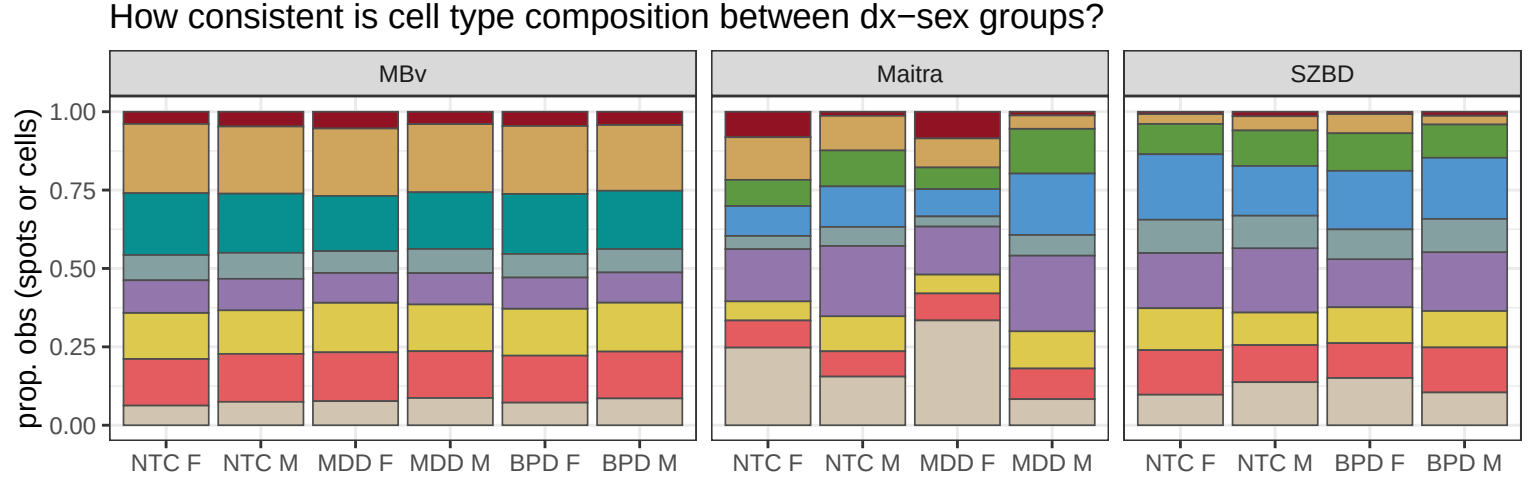

Pseudobulk sample size
M.V
Ast
L2
MBv values
for L2/3
L3
MBv values
for L2/3
L4
Inb
L5
L6
Olg
# obs (cells or spots)
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD

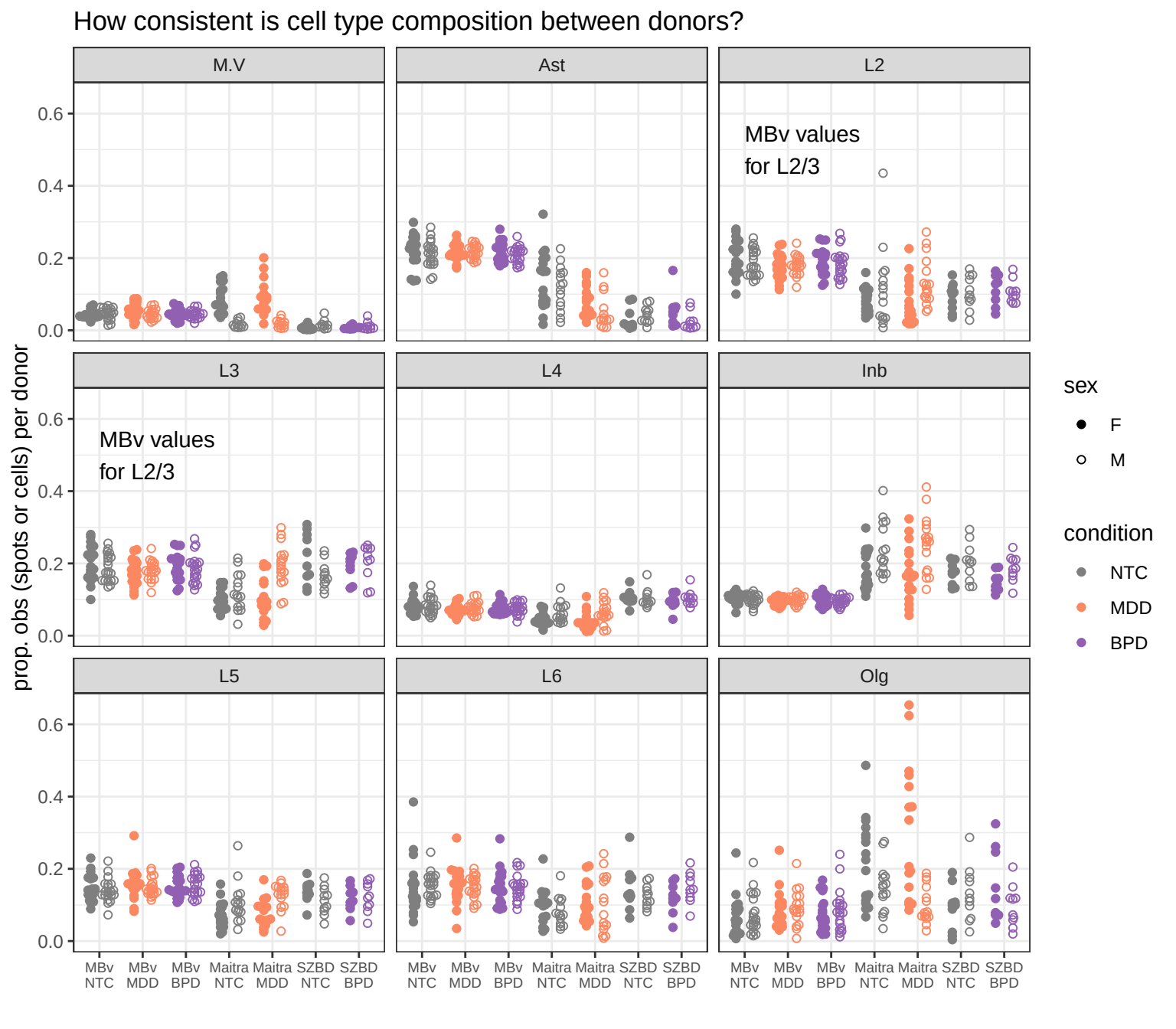

Library size (# of reads) per pseudobulk sample
M.V
Ast
L2
L3
L4
Inb
L5
L6
Olg
MBv values for L2/3
MBv values for L2/3
log10(library size)
3
4
5
6
7
8
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD

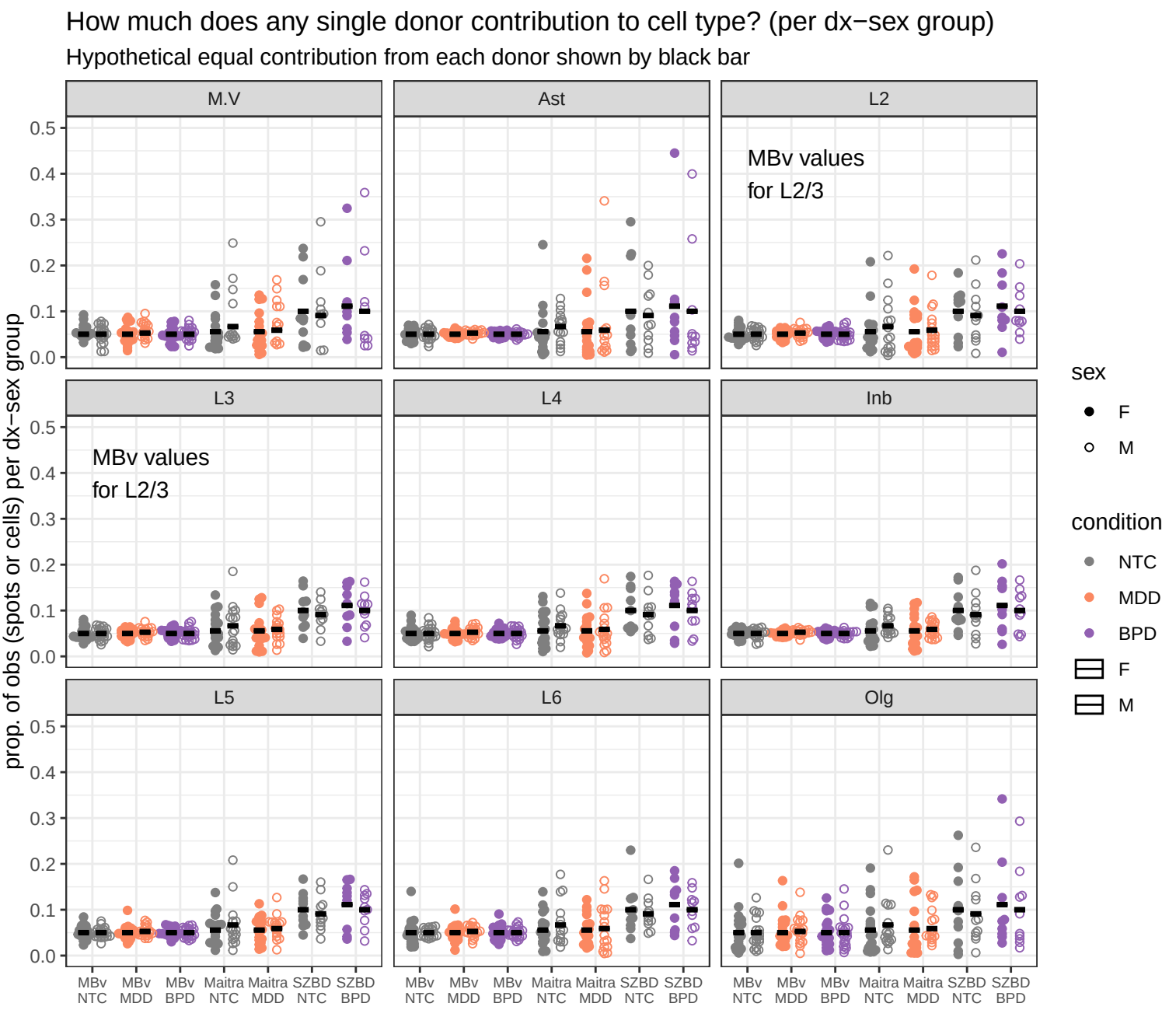

Detected genes per pseudobulk sample
M.V
Ast
L2
L3
L4
Inb
L5
L6
Olg
MBv values for L2/3
MBv values for L2/3
# unique genes
20000
15000
10000
5000
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD


Ribosomal ratio per pseudobulk sample
M.V
Ast
L2
L3
L4
Inb
L5
L6
Olg
MBv values
for L2/3
MBv values
for L2/3
percent transcripts RPS | RPL
0
5
10
15
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD


Cell type representation across dx−sex groups
MBv
Maitra
SZBD
total # obs (spots or cells)
75000
50000
25000
0
NTC F
NTC M
MDD F
MDD M
BPD F
BPD M
clus_orig
M.V
L2
L3
Inb
L6
Ast
L2.3
L4
L5
Olg


How consistent is cell type composition between dx−sex groups?
MBv
Maitra
SZBD
prop. obs (spots or cells)
1.00
0.75
0.50
0.25
0.00
NTC F
NTC M
MDD F
MDD M
BPD F
BPD M


How consistent is cell type composition between donors?
M.V
Ast
L2
MBv values
for L2/3
L3
MBv values
for L2/3
L4
Inb
L5
L6
Olg
prop. obs (spots or cells) per donor
0.0
0.2
0.4
0.6
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD


How much does any single donor contribution to cell type? (per dx−sex group)
Hypothetical equal contribution from each donor shown by black bar
M.V
Ast
L2
L3
L4
Inb
L5
L6
Olg
MBv values for L2/3
MBv values for L2/3
prop. of obs (spots or cells) per dx−sex group
0.0
0.1
0.2
0.3
0.4
0.5
MBv NTC
MBv MDD
MBv BPD
Maitra NTC
Maitra MDD
SZBD NTC
SZBD BPD
sex
F
M
condition
NTC
MDD
BPD
F
M